

Proof of Knowledge in the Data Science Degree Programme according to the Admission Regulations

Please fill out this form digitally – not by hand.

Proof of Knowledge and Expertise as stated in §2 (2) of the Admission Regulations

a) Basic knowledge, that has been acquired within the scope of the first study programme by successfully participating in equivalent study modules (courses such as lectures, exercise courses, practical courses), having achieved at least **25 credit points in basic Mathematics¹** and at least **60 credit points in basic Computer Science**, and knowledge in the following central subjects of Computer Science²:

- Programming skills
- Software Engineering
- Databases
- as well as at least 2 topics from the areas IT Security, Distributed Systems and Computer Networks

Last name	First name	Application number

Example 1: More “Computer Science background” (study programmes)

Topic	Subjects and Course Code and Credits obtained (Examples)	Total Credits (ECTS)	Courses and Credits (please fill in your completed courses and credits)
1. Mathematics	▪ Elementary Mathematics for Engineers (3) ▪ Differential Equations (3) ▪ Multivariate Analysis, Linear Algebra and Special Functions (3) ▪ Discrete Mathematics (2) ▪ Complex Analysis (2) ▪ Optimization and Calculus of Variations (3) ▪ and further....	Example: 26 (>25 ¹ ECTS in Mathematics)	CP = Credits Mathematics 1 (sem 1) - CP 5, Mathematics 2 (sem 2) - CP 4, Mathematics 3 (sem 3) - CP 4, Mathematics 4 (sem 4) - CP 4, Engineering mechanics (Sem 2) CP 4, Physics (Sem 1) CP 5
2. Programming Skills	▪ Computer Programming (2) ▪ Computer Programming Lab (2) ▪ OOP's with C++ (4) ▪ OOP's with C++ Lab (2) ▪ Programming with Python (2) ▪ Programming with Python Lab (2) ▪ Programming in JAVA (4) ▪ Programming in JAVA Lab (2) ▪ Advance Programming in JAVA (4) ▪ Adv. Programming in JAVA Lab (2) ▪ and further...	Example: 26	Programming for Problem solving (Sem 2) - CP 4, Data Structure and Algorithms (Sem 3) - CP 5 Artificial Intelligence and Neural networks (Sem 5) - CP - 4, Machine learning for Cyber security (sem 6) - CP 3 Project 1 (Sem 7) - CP 2

Topic	Subjects and Course Code and Credits obtained (Examples)	Total Credits (ECTS)	Courses and Credits (please fill in your completed courses and credits)
3. Software Engineering	▪ Software Engineering (3) ▪ Software Project Management (3) ▪ and further...	Example: 6	Software Engineering (sem 5) CP - 4, Project based learning (sem 5)CP - 1, Research based learning (Sem 6) CP -1 , Total - 6
4. Databases	▪ Database Management Systems (4) ▪ Database Management Systems Lab (2)	Example: 6	Data Base Management System (Sem 4) - CP 4
5. IT Security	▪ Network and Information Security (4) ▪ Network and Information Security Lab (2) ▪ Cryptology (3) ▪ and further...	Example: 9	Ethical Hacking in Cyber security (Sem 5) - CP 4, Information Security compliance (Sem 6)CP 4, Vulnerability assessment and penetration (Sem 6) CP 4, Application security and secure
6. Distributed Systems	▪ Distributed Systems (4) ▪ Operating Systems (4) ▪ and further...	Example: 8	Operating system and security (Sem 4) CP 4, Virtulisation and Cloud security
7. Computer Networks	▪ Computer Networks (4) ▪ Wireless Network (3) ▪ and further...	Example: 7	Computer networks (Sem 3) CP 4
8. Further Computer Science Courses	▪ Internet of Things (2) ▪ Artificial Intelligence (4) ▪ Green Computing (3) ▪ Data Mining and Data Warehousing (4) ▪ Algorithm Design and Analytics (3) ▪ and further...	Example: 16	Theory of Computation (Sem 5) CP 4, Digital System design and computer Architechture (Sem 3) CP 4 Industry Certification (Sem 7) CP 2, Industry Practice 1 (Sem 3)
Total		Example: Total: 84 (>60 ⁷² ECTS in Computer Science and Programming)	

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b) Basic knowledge, that has been acquired within the scope of the first study programme by successfully participating in equivalent study modules (courses such as lectures, exercise courses, practical courses), having achieved at least **75 credit points in basic Mathematics¹**, of which up to **15 credit points may have been achieved in Computer Science, and knowledge in the following central subjects²**:

- Linear Algebra
- Analysis, also multivariate
- Programming skills

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Example 2: More “Mathematics background” (study programmes)

Topic	Subjects and Course Code and Credits obtained (Examples)	Total Credits (ECTS)	Courses and Credits (please fill in your completed courses and credits)
1. Linear Algebra	▪ Linear Algebra (4) ▪ Advanced Linear Algebra (2) ▪ and further...	Example: 6	Mathematics I (Semester 1) – CP 4 Mathematics II (Semester 2) – CP 2
2. Analysis, also multivariate	▪ Numerical Computing and Analysis (2) ▪ Mathematical Analysis (6) ▪ Statistical Data Analysis (4) ▪ Multivariate Analysis and Nonparametric Methods (6) ▪ and further...	Example: 16	Mathematics III (Semester 3) – CP 3 Mathematics IV (Semester 4) – CP 3 Total - 6
3. Programming Skills	▪ Computer Programming (2) ▪ OOP's with C++ (4) ▪ Programming with Python (2) ▪ Programming in JAVA (4) ▪ Programming in JAVA Lab (2) ▪ Advanced Programming in JAVA Lab (2) ▪ and further...	Example: 16 (>1) → Programming skills are present!	Programming for Problem Solving (Semester 2) – CP 4 Data Structure and Algorithms (Semester 3) – CP 5 Total - 9
4. Further Math Courses	▪ Calculus (6) ▪ Stochastic Processes and Queuing (6) ▪ Discrete Mathematics (6) ▪ Numerical and Statistical Methods (2) ▪ Applied Mathematics (4) ▪ Computer Oriented Statistical Techniques Practical (4) ▪ Mathematics for Statistics (3) ▪ Differential Equations (4) ▪ Calculus and Analytical Geometry (3) ▪ Probability and Statistics (4) ▪ Graph Theory and Combinatorics (4)	Example: 46	Physics (Semester 1) – CP 5 Engineering Mechanics (Semester 2) – CP 4 Theory of Computation (Semester 5) – CP 4 Digital System Design & Computer Architecture (Semester 3) – CP 4 Total - 17

Topic	Subjects and Course Code and Credits obtained (Examples)	Total Credits (ECTS)	Courses and Credits (please fill in your completed courses and credits)
5. Computer Science	▪ Introduction to Databases (2) ▪ Introduction to Computer Science (2) ▪ Data Mining (2) ▪ Artificial Intelligence (4)	Example: 10	Artificial Intelligence & Neural Networks (Semester 5) – CP 4 Database Management Systems (Semester 4) – CP
Total		Example: 84 (>75 ¹ ECTS in Mathematics Programming) and 10 Computer Science (up to 15 ² ECTS possible) Total: 94	